HEALTHIER, LONGER, BETTER LIVES

# Medical Receipt OCR Benchmark

Traditional OCR with rules vs Azure AI extraction

Version 2 results · 9 receipts · 51 scored fields

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# Business Problem and Objective

Every medical claim needs the same few fields from a receipt; we tested how well that can be automated.

THE QUESTION

## Is Azure Document Intelligence + Azure OpenAI worth its extra cost and complexity?

Compared with open-source OCR plus fixed extraction rules that run on our own compute.

SIX FIELDS PER RECEIPT

- Doctor name
- Registration number
- Patient name
- Receipt number
- Service date
- Total amount

JUDGED ON

- Accuracy
- Speed
- Reliability
- Cost
- Maintenance effort


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# Methodology

Two steps per receipt: read the text (OCR), then find the six fields. We changed one step at a time.

| | STEP 1 · READ THE TEXT (OCR) | STEP 2 · FIND THE SIX FIELDS |
|---|---|---|
| 1 **PaddleOCR Light** | Open-source OCR, small model | Rules: original, then improved |
| 2 **PaddleOCR Heavy** | Open-source OCR, large model | Rules: original, then improved |
| 3 **Azure DI** | **Cloud OCR (prebuilt-read)** | Rules: original, then improved |
| 4 **Azure DI + AOAI** | **Cloud OCR (prebuilt-read)** | **Azure OpenAI gpt-5-mini (prompt unchanged)** |

**9 receipts**

**51 fields scored**

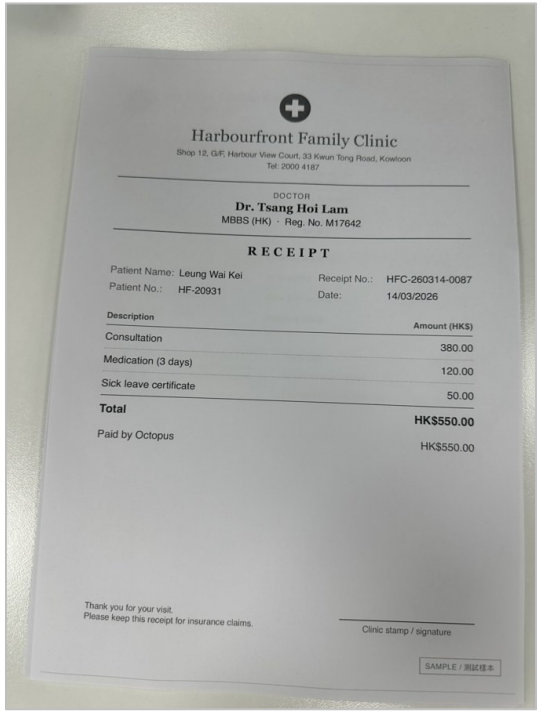
Red = paid Azure call


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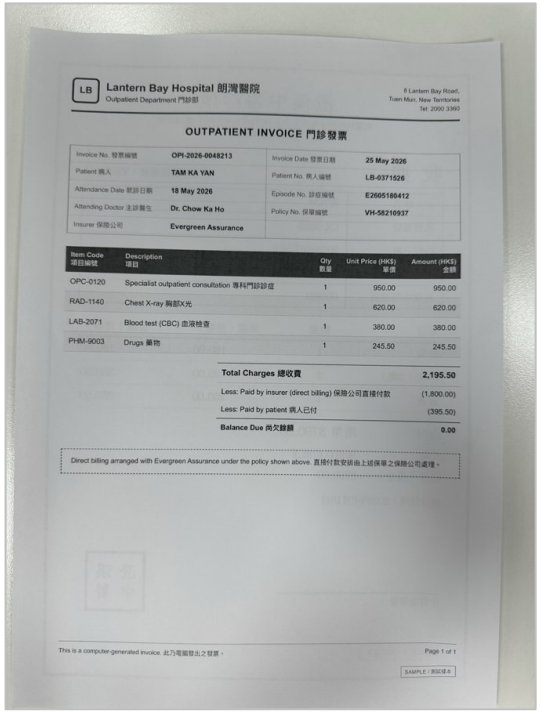
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# The Receipts: Different Layouts and Languages

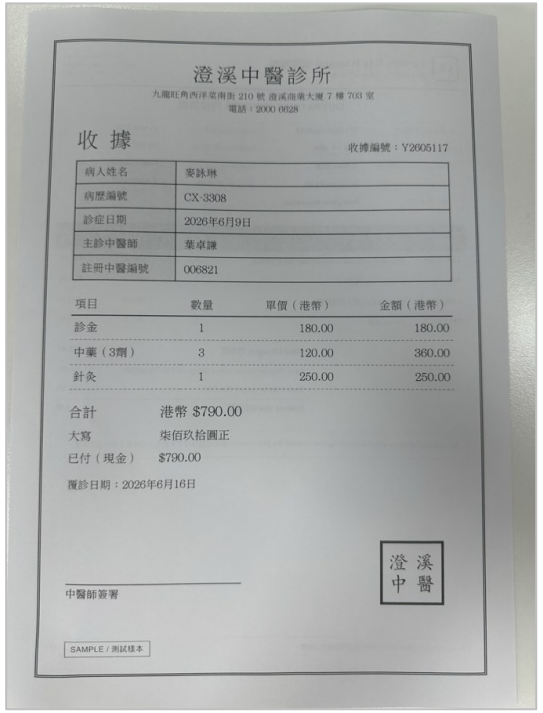
Seven new printed receipts photographed by phone, plus two receipts carried over from Version 1.




GP clinic · English




Hospital invoice · bilingual




Chinese medicine · Chinese

**ALL NINE RECEIPTS**

- English: GP, specialist, physiotherapy
- Bilingual: hospital, dental, medical centre
- Traditional Chinese: Chinese medicine
- From Version 1: vet invoice, NZ pharmacy

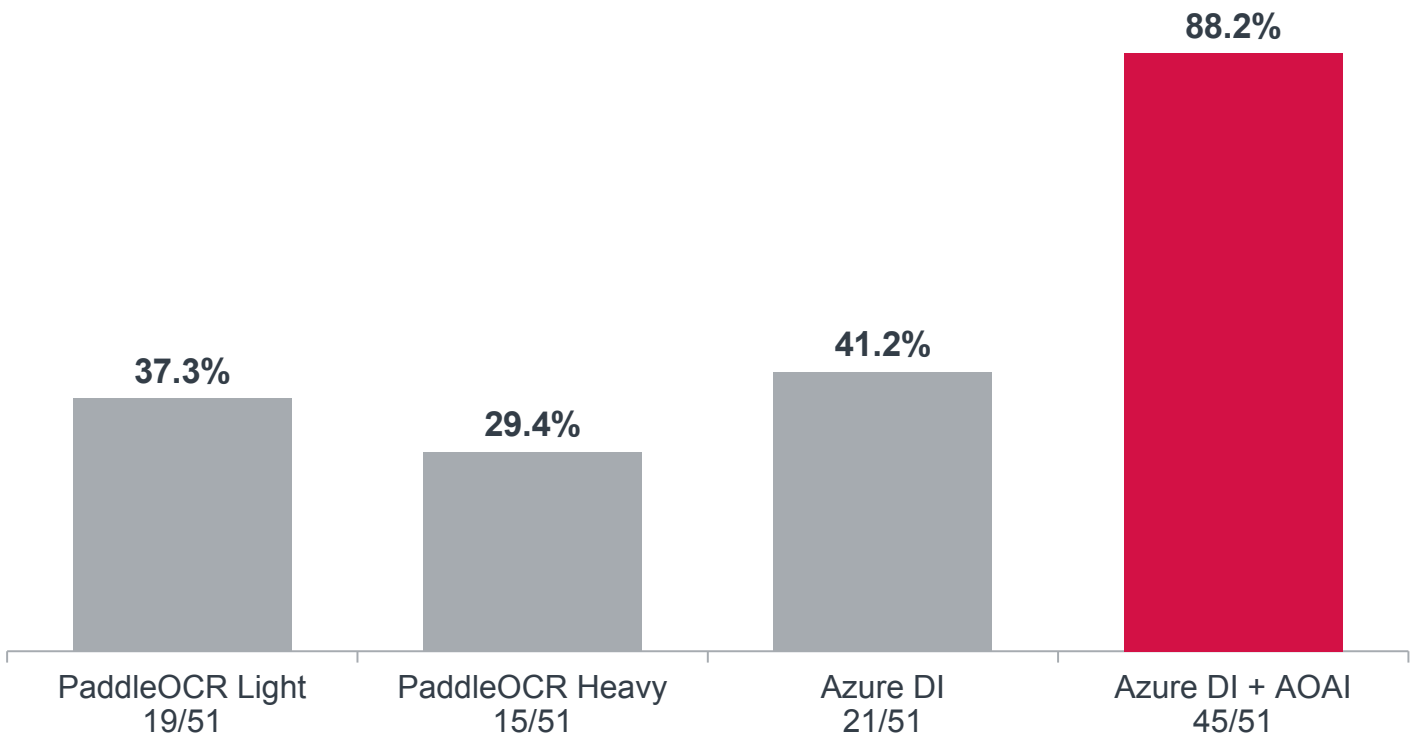
Several include distractors such as referring doctors, issue dates, patient numbers, insurer payments and balances.


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# Accuracy with the Original Rules

Azure DI + AOAI 88.2%; the three rule-based approaches 29.4–41.2%.




TWO DIFFERENT STEPS

### Text recognition (OCR)

Azure DI and PaddleOCR Light read almost every value correctly on the new receipts.

### Field extraction

The original rules only knew English labels and missed bilingual and Chinese layouts. This is where most errors came from.

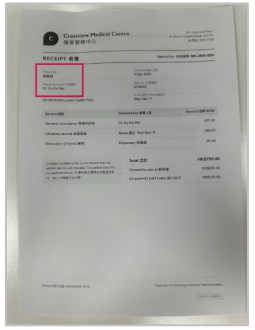
*9 receipts, 51 applicable fields. Source: python/receipt_ocr_benchmark/results/v2/scores_improved (detail.csv, summary.csv) and extraction JSON.*

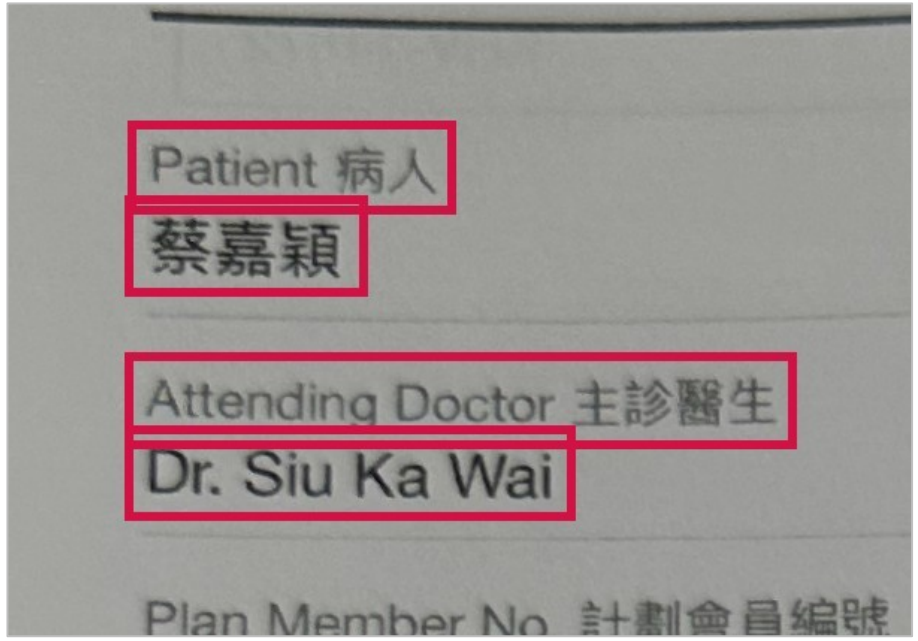

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# Example 1: Bilingual Labels (Medical Centre)

The original rules read the Chinese half of a label as the value; the improved rules and AOAI do not.






| Field | Correct value | Original rules (PaddleOCR Light) | Improved rules (PaddleOCR Light) | Azure DI + AOAI |
|---|---|---|---|---|
| **Patient** | 蔡嘉穎 | ✗ 病人 | ✓ 蔡嘉穎 | ✓ 蔡嘉穎 |
| **Doctor** | Dr. Siu Ka Wai | ✗ (not found) | ✓ Dr. Siu Ka Wai | ✓ Dr. Siu Ka Wai |
| **Reg. no.** | M19903 | ✗ (not found) | ✓ M19903 | ✓ M19903 |
| **Receipt no.** | MC-2609-3391 | ✗ (not found) | ✓ MC-2609-3391 | ✓ MC-2609-3391 |
| **Service date** | 2026-09-03 | ✗ (not found) | ✓ 2026-09-03 | ✓ 2026-09-03 |

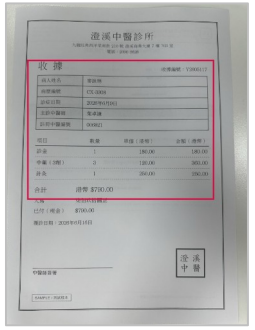
*Rules shown on PaddleOCR Light text; Azure DI with the original rules made the same patient error. ✓ correct · ✗ wrong or missing.*

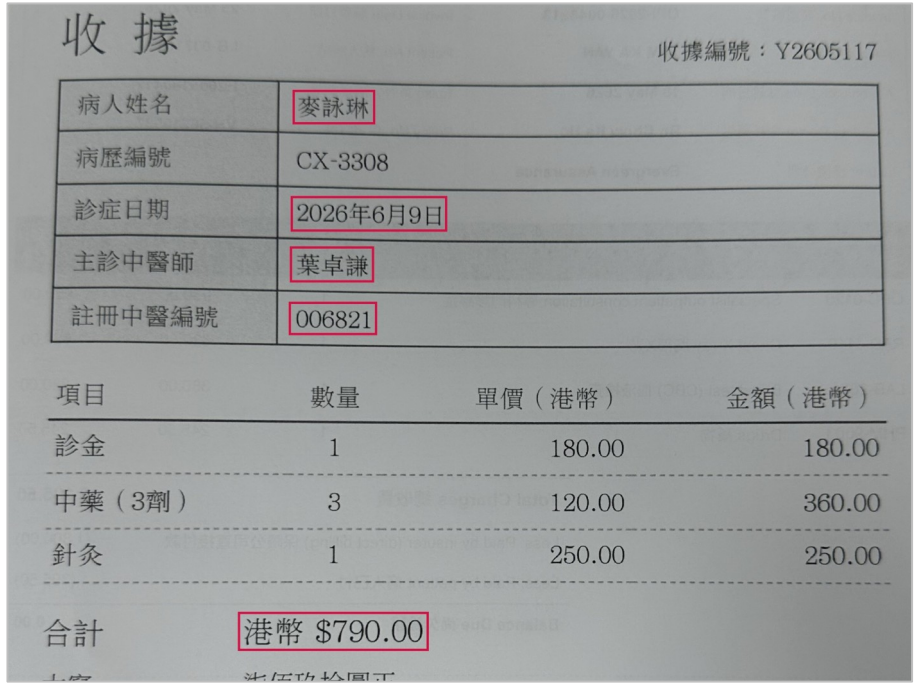

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# Example 2: Chinese-Only Receipt (Chinese Medicine)

The original rules found nothing; the improved rules and AOAI found all six fields.





收 據

收據編號：Y2605117

| | |
|---|---|
| 病人姓名 | 麥詠琳 |
| 病歷編號 | CX-3308 |
| 診症日期 | 2026年6月9日 |
| 主診中醫師 | 葉卓謙 |
| 註冊中醫編號 | 006821 |

| 項目 | 數量 | 單價（港幣） | 金額（港幣） |
|---|---|---|---|
| 診金 | 1 | 180.00 | 180.00 |
| 中藥（3劑） | 3 | 120.00 | 360.00 |
| 針灸 | 1 | 250.00 | 250.00 |
| 合計 | 港幣 $790.00 | | |

| Field | Correct value | Original rules (PaddleOCR Light) | Improved rules (PaddleOCR Light) | Azure DI + AOAI |
|---|---|---|---|---|
| **Patient** | 麥詠琳 | ✗ (not found) | ✓ 麥詠琳 | ✓ 麥詠琳 |
| **Doctor** | 葉卓謙 | ✗ (not found) | ✓ 葉卓謙 | ✓ 葉卓謙 |
| **Reg. no.** | 006821 | ✗ (not found) | ✓006821 | ✓006821 |
| **Service date** | 2026-06-09 | ✗ (not found) | ✓2026-06-09 | ✓2026 年 6 月 9 日 |
| **Total amount** | 790 | ✗ (not found) | ✓790.0 | ✓790.0 |

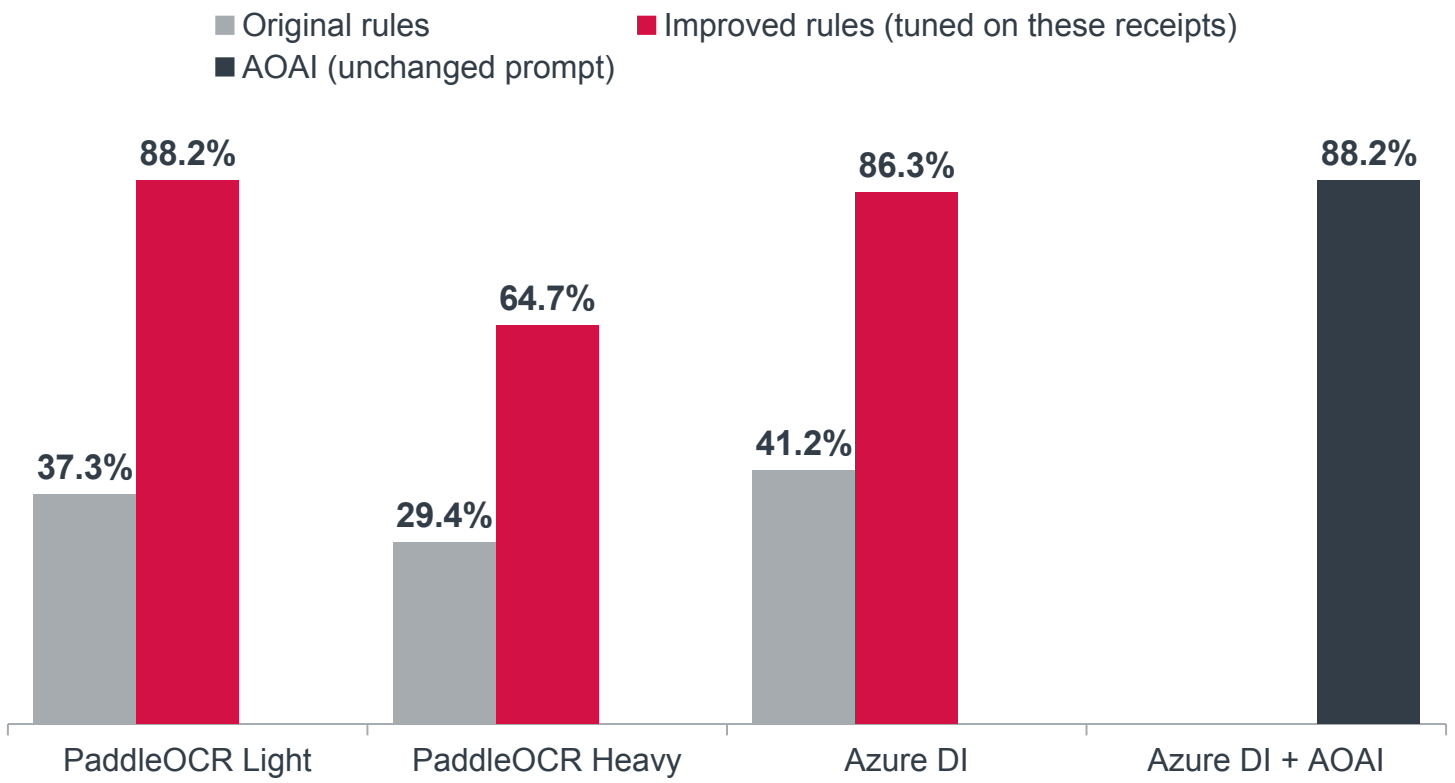
*Rules shown on PaddleOCR Light text; Azure DI and Heavy behaved the same. ✓ correct · ✗ wrong or missing.*


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# Accuracy Before and After Improving the Rules

Improved rules lifted PaddleOCR Light to 88.2% and Azure DI to 86.3% — on the receipts they were tuned on.




| 7 new receipts | Correct |
|---|---|
| PaddleOCR Light (improved) | **41/41** |
| Azure DI (improved) | **40/41** |
| PaddleOCR Heavy (improved) | **29/41** |
| Azure DI + AOAI | **39/41** |

**Read with care**

Equal scores here do not mean equal accuracy on unfamiliar receipts: the rules were tuned on these receipts, the AOAI prompt was not.

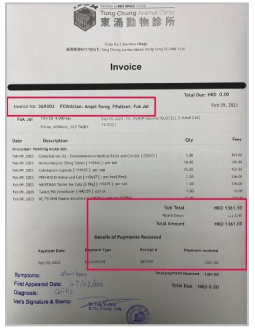
*51 applicable fields per approach. Source: python/receipt_ocr_benchmark/results/v2/scores_improved (detail.csv, summary.csv) and extraction JSON.*

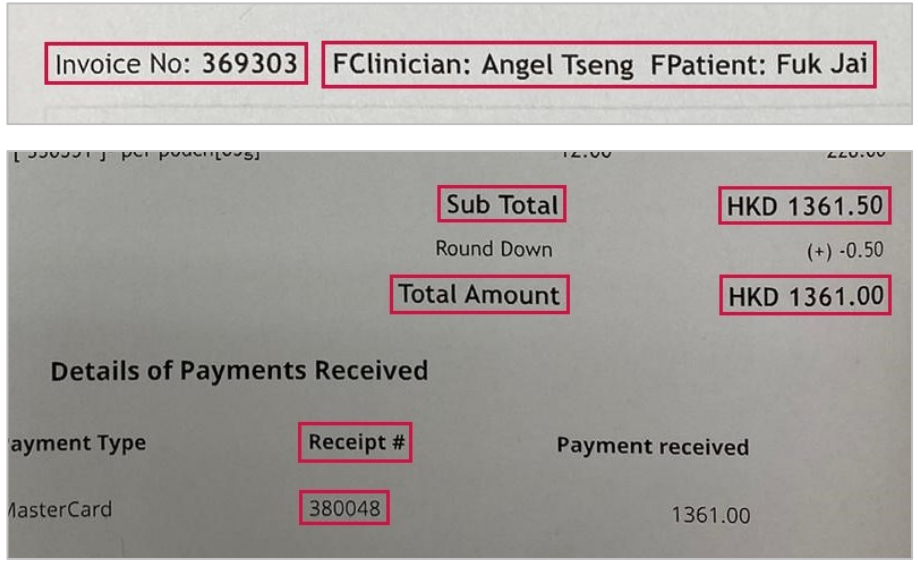

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# Example 3: What Still Goes Wrong (Vet Invoice)

Each approach fails differently: a rule change picked the wrong number, AOAI picked the subtotal.






| Field | Correct value | Original rules (PaddleOCR Light) | Improved rules (PaddleOCR Light) | Azure DI + AOAI |
|---|---|---|---|---|
| **Receipt no.** | 369303 | ✓ 369303 | ✗ 380048 | ✓ 369303 |
| **Patient** | Fuk Jai | ✗ (not found) | ✗ (not found) | ✓ Fuk Jai |
| **Doctor** | Angel Tseng, Lok Yi Choi | ✗ (not found) | ✗ (not found) | ✗ Angel Tseng |
| **Total amount** | 1361 | ✓ 1361.0 | ✓ 1361.0 | ✗ 1361.5 |

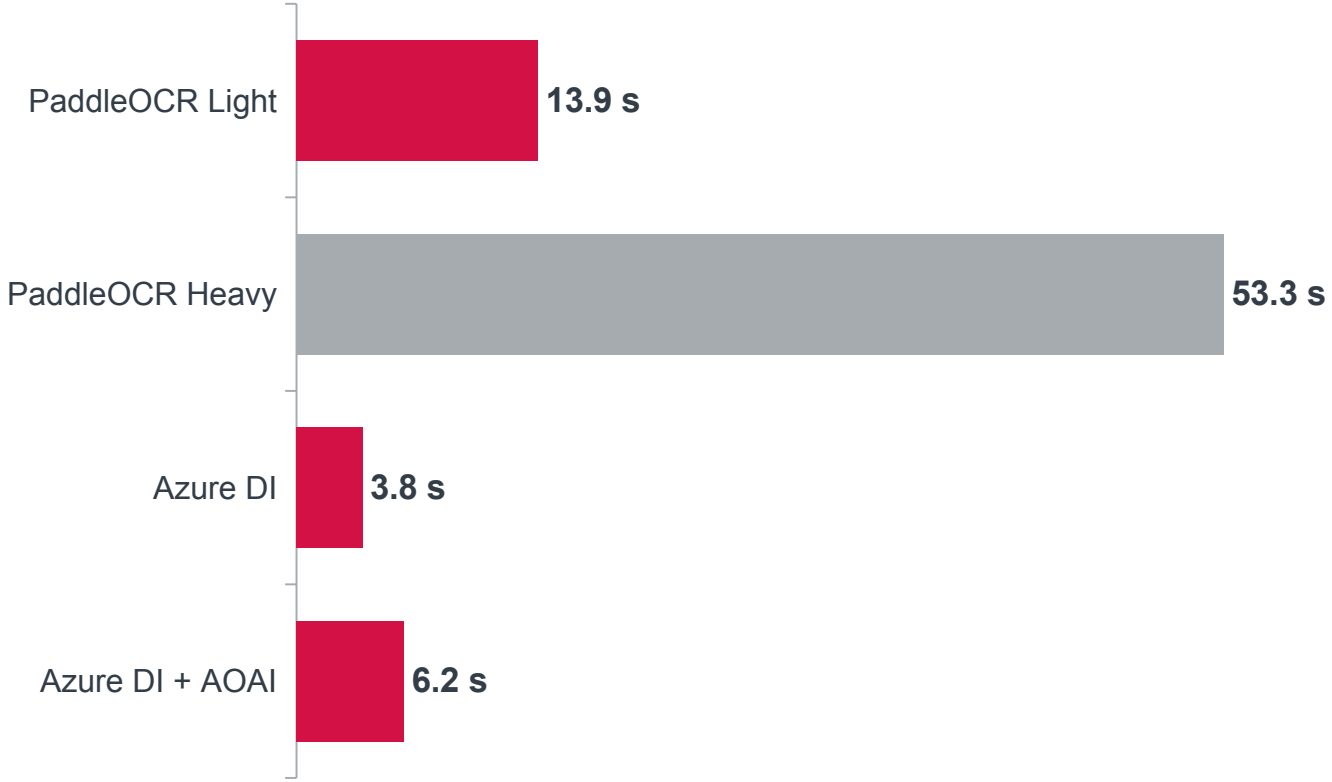
*PaddleOCR Heavy also found almost no text on two photos (3 and 0 text blocks): an OCR failure no rule can fix. ✓ correct · ✗ wrong or missing.*


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# Speed, Reliability and Cost

Azure DI is fastest (3.8 s); every approach processed all 9 receipts.




*Times measure different things (CPU vs API call incl. network); receipts were processed one at a time.*

**RELIABILITY**

**9 / 9**

receipts processed by every approach, no failed requests

**COST DRIVERS (MEASURED)**

- Rules: 4–10 ms of CPU per receipt
- PaddleOCR: CPU time on our compute; no per-call fee
- AOAI: about 602 input + 263 output tokens per receipt

**Prices still to be confirmed**


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# Key Findings and Trade-offs

Most errors came from field extraction, not OCR; better rules close the gap on known layouts.

1. **OCR was mostly right**
   On the new receipts, Azure DI and PaddleOCR Light read almost every value.

2. **Original rules were weak**
   English-only labels: 37.3–41.2% for Light and DI.

3. **Rules can be improved**
   Tuned rules: Light 88.2%, DI 86.3%; label vocabulary grew about 3×.

4. **AOAI handled new layouts as-is**
   88.2% with a prompt never tuned to these receipts.

5. **Heavy adds nothing**
   Lowest accuracy and slowest (53.3 s); OCR failed on two photos.

| | PaddleOCR Light + improved rules | Azure DI + AOAI |
|---|---|---|
| **Runs where** | Our compute, offline | Azure cloud services |
| **Per-request fee** | None (compute only) | DI pages + AOAI tokens |
| **New layouts** | Needs new rules | Handled by the prompt |
| **Main risk** | Rule maintenance | Cost, model variability |


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# Recommendation and Next Steps

Keep two candidates and decide on a larger set of receipts that neither has seen.

CANDIDATE 1

**PaddleOCR Light + improved rules** **88.2% (tuned) · 13.9 s**

Offline, no per-request charges; needs rules maintained for each new layout.

CANDIDATE 2

**Azure DI + AOAI** **88.2% · 6.2 s**

Flexible on varied layouts with an unchanged prompt; API costs and run-to-run variability.

**Limitation:** 9 receipts, mostly mock and printed; the improved rules were tuned on them.

NEXT STEP 1

### Freeze both

Lock the improved rules and the AOAI prompt before any new test

NEXT STEP 2

### Test on unseen receipts

30+ real, anonymised receipts; count rule fixes needed per new layout

NEXT STEP 3

### Confirm prices

Compute rate, Document Intelligence pages, gpt-5-mini tokens


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